Shiqi HE

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EDUCATION

University of Michigan (UMich) **09 2023 – Present**
PhD in Computer Science and Engineering *Ann Arbor, USA*
- Area(s): Agentic AI Systems, Efficient ML Systems, Multimodal Model Training
- Graduate Student Instructor: CSE 585 Advanced Scalable Systems for Agentic AI
- Advisor: Mosharaf Chowdhury

University of British Columbia (UBC) **09 2020 – 05 2023**
Master of Science (MSc) in Computer Science - CGPA - 88.8/100 *Vancouver, Canada*
- Area(s): Federated Learning
- Advisors: Ivan Beschastnikh and Mathias Lécuyer

Hong Kong Polytechnic University (PolyU) **09 2016 – 08 2020**
BSc (HONS) Computing, Department of Computing - CGPA - 3.90/4.00 *Hong Kong SAR*
- Outstanding Student Award (1 / 150)
- Advisor: Song Guo

Korea Advanced Institute of Science and Technology (KAIST) **02 2019 – 06 2019**
Exchange Program, School of Computing - CGPA - 3.60/4.30 *Daejeon, Korea*

SELECTED PUBLICATIONS & PREPRINTS

[1] Jiaheng Lu, **Shiqi He**, Xian Wang, Xuting Liu, Zhuoming Chen, Chuyue Wang, Jiaming Liang, Haizhong Zheng, Linh Phan, Mosharaf Chowdhury, Beidi Chen, and Vincent Liu. "BrowSA: Semantic Aware Sparse Attention for Web Agents". In: *In submission* (2026).

[2] **Shiqi He**, Yue Cui, Feijie Wu, Xinyu Ma, Jiaheng Lu, Yaliang Li, Bolin Ding, and Mosharaf Chowdhury. "Beyond Domains: Reusing Web Skills via Transferable Interaction Patterns". In: *Conference on Neural Information Processing Systems (NeurIPS 2026)*. 2026.

[3] **Shiqi He**, Yue Cui, Xinyu Ma, Yaliang Li, Bolin Ding, and Mosharaf Chowdhury. "Branch-and-Browse: Efficient and Controllable Web Exploration with Tree-Structured Reasoning and Action Memory". In: *Annual Meeting of the Association for Computational Linguistics (ACL 2026)*. 2026.

[4] **Shiqi He**, Insu Jang, and Mosharaf Chowdhury. "Mordal: Automated Pretrained Model Selection for Vision Language Models". In: *International Conference on Learning Representations (ICLR 2026)*. 2026.

[5] Runyu Lu, **Shiqi He**, Wenxuan Tan, Shenggui Li, Ruofan Wu, Jeff J Ma, Ang Chen, and Mosharaf Chowdhury. "TetriServe: Efficiently Serving Mixed DiT Workloads". In: *ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS 2026)*. 2026.

[6] Technical Report. "AgentScope 1.0: A Developer-Centric Framework for Building Agentic Applications". In: *arXiv preprint arXiv:2508.16279* (2025).

[7] Zhifeng Jiang, Peng Ye, **Shiqi He**, Wei Wang, Ruichuan Chen, and Bo Li. "Lotto: Secure Participant Selection against Adversarial Servers in Federated Learning". In: *USENIX Security Symposium 2024*. 2024.

[8] **Shiqi He**, Qifan Yan, Feijie Wu, Lanjun Wang, Mathias Lécuyer, and Ivan Beschastnikh. "GlueFL: Reconciling Client Sampling and Model Masking for Bandwidth Efficient Federated Learning". In: *Conference on Machine Learning and Systems (MLSys 2023)*. 2023.

[9] Feijie Wu, **Shiqi He**, Song Guo, Zhihao Qu, Haozhao Wang, Weihua Zhuang, and Jie Zhang. "Sign Bit is Enough: A Learning Synchronization Framework for Multi-hop All-reduce with Ultimate Compression". In: *ACM/IEEE Design Automation Conference (DAC 2022)*. 2022.

[10] Rindranirina Ramamonjison, Timothy Yu, Raymond Li, Haley Li, Giuseppe Carenini, Bissan Ghaddar, **Shiqi He**, Mahdi Mostajabdaveh, Amin Banitalebi-Dehkordi, Zirui Zhou, et al. "NL4Opt Competition: Formulating Optimization Problems Based on Their Natural Language Descriptions". In: *Conference on Neural Information Processing Systems (NeurIPS 2022) Competition Track*. 2022.

WORK EXPERIENCE

Adobe **05 2026 – 08 2026**
Research Intern *San Jose, USA*

- Efficient Virtual Machine Sandboxes for Computer-Use Agents
  - Running full desktops with professional software in VM-level sandboxes for computer-use agents
  - Designing a copy-on-write fork primitive that shares a frozen memory template across sandboxes
  - Developing lifecycle-aware scheduling for multi-sandbox fleets that staggers the bursty compute phases of concurrent rollouts to relieve CPU contention at scale

Alibaba **05 2025 – 08 2025**
Research Intern, Tongyi Lab *Hangzhou, China*

- AgentScope 1.0: A Developer-Centric Framework for Building Agentic Applications
  - Enabling controllable web agent exploration with tree-structured reasoning
  - Storing and reusing past actions via action memory to reduce repeated interactions

Hong Kong University of Science and Technology (HKUST) **06 2023 – 08 2023**
Research Assistant, Supervised by Prof. Wei Wang *Hong Kong SAR*

- (USENIX Security 2024) Lotto: Secure Participant Selection against Adversarial Servers in Federated Learning
  - Enabling verifiable randomness for random selection
  - Approximating informed selection through a refined process

Huawei Canada **06 2022 – 11 2022**
Support Researcher (Part-Time), Big Data and Intelligence Platform Lab *Vancouver, Canada*

- (NeurIPS 2022 Competition) NL4Opt: Natural Language for Optimization Modeling
  - Formulating optimization problems based on their natural language descriptions
  - Helped host the competition (total prize pool of $22,000)

Hong Kong Polytechnic University (PolyU) **06 2021 – 09 2021**
Research Assistant, Edge Intelligence Lab, Supervised by Prof. Song Guo *Hong Kong SAR*

- (DAC 2022) Sign Bit is Enough: A Learning Synchronization Framework for Multi-hop All-reduce with Ultimate Compression
  - A sign-bit compression-based learning synchronization framework for decentralized machine learning
  - Attended the conference in San Francisco for oral presentation

Microsoft Research Asia (MSRA) **06 2019 – 09 2019**
Research Intern, System Research Group, Supervised by Dr. Yang Chen *Beijing, China*

- (Microsoft Hackathon 2019 Beijing) Confirmation Time Prediction in Ethereum
- (SOSP 2021) Forerunner: Constraint-based Speculative Transaction Execution for Ethereum
  - A computing framework to accelerate Ethereum transaction processing through speculative execution

Hong Kong Polytechnic University (PolyU) **01 2019 – 03 2019**
Research Assistant, Alibaba-PolyU Research Group, Supervised by Prof. Jiannong Cao *Hong Kong SAR*

- Food Traceability System: A Web App based on Hyperledger Fabric (Node.js SDK) for food traceability
  - Conducted performance testing and debugging for the system and the Ionic frontend interface
  - Assisted in setting up Hyperledger Fabric service in multi-node using Docker and debugging Fabric smart contract

OTHER PROJECTS

Graduate Course - Security and Privacy | Term Paper **2021**
"An Analysis of Existing Spectre Mitigations in eBPF" *UBC*

Graduate Course - Distributed System | Term Paper **2019**
"Time-Bounded API Energy Minimization Offloading Strategy in Mobile OCR Applications" *KAIST*

AWARDS & SCHOLARSHIPS

HKSAR Government Scholarship Fund - Talent Development Scholarship	Scholarship	**2020**
The Hong Kong Polytechnic University Scholarship	Scholarship	**2019**
Microsoft Research Asia Stars of Tomorrow Internship Program	Award of Excellence	**2019**
Bright Future Charitable Foundation Asian Exchange Scholarship	Scholarship	**2019**
2018 ACM-ICPC Asia Regional Contest Shenyang Site	Silver Medal	**2018**
2018 EY (Ernst & Young) Cyber Asia Case Competition	First Runner-up Team (HK$10,000)	**2018**
24-Hour IEEE Xtreme Programming Competition 11.0	World Top 30 (3%)	**2017**
PolyU COMP Freshman Project Competitions	First Runner-up Team	**2016**
China National Olympiad in Informatics in Provinces (NOIP)	First Prize (Hunan Province)	**2014**